

zh3r0 2020 Crypto - Uncipher Me

deuterium — 2020-06-18

Canonical HTML: [https://deut-erium.github.io/WriteUps/2020/zh3r0/crypto/Uncipher`me/2020-06-18-zh3r0-2020-Uncipher-Me.html](https://deut-erium.github.io/WriteUps/2020/zh3r0/crypto/Uncipher%27me/2020-06-18-zh3r0-2020-Uncipher-Me.html)

Abstract

CTF writeup: zh3r0 CTF 2020's Uncipher Me sends us on an encoded-key scavenger hunt. The ROT47-derived key opens the Fernet token; the other keys are busywork.

Uncipher Me

Description

Challenge Files

- [ct.txt](#)
- [key.txt](#)
- [key`2.txt](#)
- [key`3.txt](#)

Lets begin with the keys.

```
cat key.txt
TjogR0kzRE90WldHUVpER01SVkdFWURLT0JWSEUyRE1NSl1HNfPEQU5CvEdBNFRJTUJSR0VZVEdNQLpHTTRUU0
1CWUdJmLRNTUJRR1U0REl0SlhHUTJUS05CUUdBM1RLTlpTRzQzRE1PQlRIRTJESU9KWUc0M0RRTVpXSEU0VE1N
WlpHUVlUU01CvEHfWkRBTkpYSEU0RE1NSldHTVlURU5KU0c0NFRDTPVpVSEU0REl0SlFHUVlUTU5SV0dNNFRBTv
JYR000VFF0UlVHsvLUQ01SWkdJmKRJTkJYR0EzVFNNULZHSTRE0Q05aWUc0NERBTlpVR1LZRE1NQLRHUTNEU9C
WEc0WkRHTkpWSEVZRFN0QLJIQTJUS01KV0dZNFRT0pVR1EzVEtPQlRHNDRTU09CVkdJNERPTUPRR1VaVE10QL
VHRTRET05SWkdBWVRTT0JURzQ0REFNWlRHRVpUQ09KVUdRM0RJTLpTR1FaRE90UlRHWTNES09KVEdFwKRHT0JW
R0UzVEVPSlNIQTRUTU5SVkdNWURTTVJXR0EyRE90SlVHNDRTU01CWEdFWURHTVJYR1VaVE1PSlZHNfLEUU5KVk
hBNERNTlJSR01aVFF0UlDhSTNET05aU0dBM0RDT0pSR00zREF0UlRHWTJUU01KVUhBNERPTlpSR0kzREt0WlJH
WTJEQ09KWUdFM0RTTlpRR0LZRFF0WlJHUVpETU1CvEHbWkRBTkJYSEU0VENNSlpHQTNuQU1SVehFWlRBTUJWR0
FaREN0WlVHWTJUSU1aU0c0WVRRTVpXR0U0RFFPSlhHQVpUU09CU0dZNFRT0JUR00yVEd0QLpHSTJERU1CUkc0
NFRHTVpSSEUzREt0WlZHUTRETU5KWkdJmKRRTkJYSEezRENOQLpHQTREUR1aVEdZWVRFTkJZR0U0VEVNULFHTT
RET01CVkdZWVRLTUpSR01aVEVPQlJHSTJUUU9CWEdVMlRLTkjZR0LZRE10QLNHQVlER01KWEhBwlrFTVJTR0Uz
VFFPQlNHRTRETU5KWUdBMlRPTUpUR0VaVEV0UlHHTRETU1KUUDJNERJTUpZRzRZVE9PSlVIRTRET09CUUdJWL
RNTUpXR00yVEVPQlHSTJUT09KU0dNM1RRT0pSRzRaVeLNUlNHSVlUT09KUKdRWVRJTkpZRzQzVEdPQlJHwVpU
R09CU0hBM0RRTlpZR1FaVFNOQLpHWTNEQU1aU0dVM1RHTkJSR1FaVENPQlRHUTNUTU1aVUdRMKRHTVpTR0FZRE
FPSlJHSTJEU01KU0dVM0RJTLpUSEVaUT09PT0KIGU6IEdZMLRLTVpYCiBjdDogIEdCN2tUSGFSZGtII0LSZkha
P1JqSDgoYWZHYy0wZUcmZUXiSTVJUmlIYUlwbEcmV9tR2N6K2VGZj0oWkYpPWhZSDhEOWRIODM/Wkgj0WduST
U5TGfJV3tzZUdjLTlpR0JQKfLHQjdZV0LXYWfM5VdzGdIYTbqbkcl9nSVhFfm1I0TBnakLXI3lvSVhF02LH
ZERSa0k1anNwSCNqdmPJV2FSY0djXzxaR2RWYWhI0D9qbUdCITBiSCE/RmZHJmVVbUYqIUNKSCM5VWpIWj9GZU
ZmbGhhRmZjZVdHY3o8Ykk1UmdnSCE/T2ZiWndJZ0k1YVhhSDgzX2VHJjRDbEgJUlhmSCNJZGLGZ1A/WUgjYXNr
R0JHemFGZ1EzZkk1e35vRipQPGJGKLb0VEhaZTLnSTVhcGtJvYnwbkhaVXxaSVhPNW1IYUltaEhaZF9YR0JQdF
VGZmxZVklXI3lxRipp0WVIODMrWEgJYXBM2NgNmdGZzdYkh0XBpSfo/TGdHZFZkakdCaCtkSFpLM2dIOTBk
a0YqcjZpSVdhTgdJV3smbUlXYVJiSTVzJnJJNVJka0LXYUxnSGfJZGNHQLAoWUcmblJqSCNJZ2LHQLAkZEdjYE
ZqRyZuUmZHQmAzwUK10ztsSFpuRmZHJSsrVkyPfwJTSCM5VWpGZ1o5ZkdCK3xYRyZESWhHY3ErZkhh0VJoRmdQ
JFhJWE47Z0dCUChiRiklVLZGZn13ZEZmJWJWSCM5UmdIIzLhaUhhSWplSTVqewtHYy0zakclKz9YSCE/SwhHJV
8/WUYqITZjRil9ZVdJNTLYaELYRSNpSTVqZ2ZGKiEzaEghdl9XR0I3elpJNTAzV0YpPW5ZSfooRmJIWmR8VklX
```

```
02dmRmZ9dGNGZ0d3VkyqUChISGE5cG1HZE02Y0k1SWdmR0JoPFpHZFZJZkYqUChjSCNzJm1HQit8YUYqaHxiSF
pVK1pGKiE2ZUdkVxrsVgl3RHZE1YbkhaKExoSCNJT2RJNTAzWEZmPwhVRzU=
```

Seems like base64

```
import base64
key1 = base64.b64decode(key1)
print(key1.decode())

# N:
GI3DONZWGQZDGMVRGGEYDK0BVHE2DMMJYG4ZDANBTGA4TIMBRGEYTGMBZGM4TSMBYGI2TMMBQGU4DINJXGQ2TKN
BQGA3TKNZSG43DMOBTHE2DIOJYG43DQMZWHE4TMMZZGQYTSMBTHEZDANJXHE4DMMJWGMYTENJSG44TCMZUHE4D
INJQGQYTMNRWGM4TAMRXGM4TQNRUGIYTCMRZGI2DINBXGA3TSMRVGI4DCNZYG44DANZUGYYDMMBTGQ3DAOBXG4
ZDGNJVHEYDSNRHA2TKMJWGY4T00JUGQ3TK0BTG44T00BVGI4DOMJQGUZTMNBUGE4DONRZGAYTS0BTG44DAMZT
GEZTC0JUGQ3DINZSGQZDONRTGY3DK0JTGEZDGOBVG3TE0JSHA4TMNRVGMYSMRWGA2DONJUG44TOMBXGEYDGM
RXGUZTM0JVG4YDQNJVHA4DMNRRGMZTQNRWGI3DONZSGA3DC0JRG3DANRTGY2TSMJUHA4DONZRG3DKNZRGY2D
C0JYGE3DSNZQGIYDQNZRGQZDMMBTHAZDANBXHE4TCMJZGA3TAMRTHEZTAMBVGAZDCNZUGY2TIMZSG4YTQMZWGE
4DQ0JXGAZTS0BSGY4TA0BTGM2TGNBZGI2DEMBRG44TGMZRHE3DKNZVGQ4DMNJZGI2DQNBXHA3DCNBZGA4DEMZT
GYYTENBYGE4TEMRQGM4DOMBVGYTKMJRGMZTE0BRGI2TQ0BXGU2TKNBYGIYDMNBSGAYDGMJXHAZTEMRSGE3TQ0
BSGE4DMNJYGA2T0MJTGEZTENRYGM4DMMJQGI4DIMJYG4YT00JUHE4D00BQGI2TMMJWGM2TE0BXGI2T00JSGM3T
Q0JRG4ZTIMRSGIYT00JRGQYTIJY43TG0BRGYZTG0BSHA3DQNZYGQZTSNBZGY3DAMZSGU3TGNBRGQZTC0BTGQ
3TMMZUGQ2DGMZSGAYDA0JRG3DSMJSGU3DINZTHEZQ====

# e: GY2TKMZx
# ct: GB7kTHaRdkH#IRfHZ?RjH8(afGc-0eG&eLbI5IRiHaIplG&V0mGcz+eFf=(ZF)=hYH8D9dH83?
ZH#9gnI59LaIW{seGc-
9iGBP(YGB7YWIWaaFIWspgHa0jnG%z_gIXE~mH90gjIW#yoIXE;iGdDRkI5jspH#jvjIWARcGc_<ZGdVahH8?
jmGB!0bH! ?FfG&eUmF*!CdH#9UjHZ?FeFflhaFfceWGcz<bI5RggH! ?
0fHZwIgI5aXaH83_eG&4ClH#RXfH#IdiFgP?
YH#askGBGzaFgQ3fI5{~oF*P<bF*PtTHZe9gI5apkIW#pnHZU|ZIX05mHaImhHZd_XGBPtUFflYVIW#yqF*i9e
H83+XH#apfGc`6gFg7qbHa9piHZ?
LgGdVdjGBh+dHZe3gH90dkF*r6iIWaLgIW{&mIWaRbI5s&rI5RdkIWaLgHaIdcGBP(YG&nRjH#IgiGBP$dGc`F
jG&nRfGB`3YI5;;LHZnFfG%++VF)}bSH#9UjFgZ9fGB+|XG&DIhGcq+fHa9RhFgP$XIXN;gGBP(bF)%VVFf}wd
Ff%bVH#9RgH#9aiHaIjeI5jykGc-3jG%+?XH!?IhG%_?
YF*!6cF)}eWI59XhIXE#iI5jgFf*!3hH!v_WGB7zZI503WF)=nYHZ(FbHZd|VIW;gFFf}tcFgGwVF*P(bHa9pm
GdM6cI5IgfGBh<ZGdVIfF*P(cH#s&mGB+|aF*h|bHZU+ZF*!6eGdVLkIX5{tGdMXnHZ(LhH#I0dI503Xff=hUG
5
```

Alright!, Seems like RSA inside the key huh

N and e seem to be base32, since they are all CAPS ct appears to be base85, since all these weird characters

```
N = key1[1]
e = key1[3]
c = key1[5]

N = int(base64.b32decode(N))
e = int(base64.b32decode(e))
c = int(base64.b85decode(c))
p,q = sympy.ntheory.primefactors(N)
d = pow(e, -1, (p-1)*(q-1))
m = pow(c,d,N)

m = bytes.fromhex(hex(m)[2:])
print(m.decode())
```

This brings us

```
key: kD87Ef4y043nhee-W_Hytc0d3Bkiw4Gw21m-7AHpZkc=
This is one of the keys for the encryption. Totally 3 keys to decrypt the encrypted
text
```

Okay, time for round 2

```
cat key_2.txt
Are your rots strong.

<6Ji : "+:;v5@)_96AGaH?u+~&D%(&\GeIJv(JB$2?0Afdrtl
w6C6VD 2?@E96C <6J 7@C E96 6?4CJAE:@?]
:7 J@F 2C6 2 8@q5 4CJAE@8C2A96C J@F 42? :56?E:7J E96 4@>@? DJ>>6EC:4 6?4CJAE:@?]
```

Seems like a much anticipated rot47, dcode.fr to our rescue

The screenshot shows the dCode.fr website interface for the ROT-47 cipher tool. The page is in French, with a 'Lire en Français' link in the top right. The main content area is titled 'ROT-47 CIPHER' and includes a 'Summary' section with a list of links to related tools and concepts. The 'Similar tools' section lists various other encryption tools like Caesar Cipher, ROT Cipher, and Enigma Machine. The 'Support' section lists links to PayPal, Patreon, and a 'More' link. The 'Answers to Questions' section provides a definition of the ROT-47 cipher and a conversion table for the alphabet.

ROT-47 CIPHER
Cryptography › Substitution Cipher › ROT-47 Cipher
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ROT47 DECODER
★ ROT47 CIPHERTEXT
<6Ji : "+:;v5@)_96AGaH?u+~&D%(&\GeIJv(JB\$2?0Afdrtl
w6C6VD 2?@E96C <6J 7@C E96 6?4CJAE:@?]
:7 J@F 2C6 2 8@q5 4CJAE@8C2A96C J@F 42? :56?E:7J E96 4@>@? DJ>>6EC:4 6?4CJAE:@?]
[DECRYPT ROT47]

See also: [ROT Cipher](#) – [ROT-13 Cipher](#) – [Caesar Cipher](#)

ROT47 ENCODER
★ CAESAR CODE PLAIN TEXT
dCode Rot-47
[ENCRYPT WITH ROT-47]

See also: [ROT Cipher](#) – [ROT-13 Cipher](#) – [Caesar Cipher](#)

Tool to decrypt/ encrypt with Rot47. The ROT-47 cipher is a variant of the ROT-13 suitable for ASCII characters, exactly a subset of 94 printable characters.

ANSWERS TO QUESTIONS
What is Rot-47? (Definition)
The **Rot-47** is a shift cipher that improves the **Rot-13** by allowing it to encode almost all visible **ASCII** characters (where **Rot13** could only encode letters).
To achieve this, **Rot47** uses a 94-character alphabet that is a subset of the **ASCII** characters between the character 33 ! and the character 126 ~.
How to encrypt using Rot-47?
Rot47 consists in replacing a character with another located 47 positions **after in the alphabet**. The conversion table is:
! " # \$ % & ' () * + , - . / 0 1 2 3 4 5 6 7 8 9 : ; < = > ? @ A B C D E F G H I J K L M N O P Q R S T U V W X Y Z [\] ^ _ ` a b c d e f g h i j k l m n o p q r s t u v w x y z { | } ~

Summary
★ [ROT47 Decoder](#)
★ [ROT47 Encoder](#)
★ [What is Rot-47? \(Definition\)](#)
★ [How to encrypt using Rot-47?](#)
★ [How to decrypt Rot-47 cipher?](#)
★ [Why the shift of 47?](#)
★ [How to recognize ROT-47 ciphertext?](#)
★ [What are the variants of the Rot-47 cipher?](#)
★ [What is the algorithm of Rot 47? \(Pseudo-code\)](#)

Similar tools
★ [Caesar Cipher](#)
★ [ROT Cipher](#)
★ [ROT-13 Cipher](#)
★ [Letter Number \(A1Z26\) A=1, B=2, C=3](#)
★ [Morse Code](#)
★ [Mono-alphabetic Substitution](#)
★ [Pseudonyms/Nicknames Generator](#)
★ [Binary Code](#)
★ [Polybius Cipher](#)
★ [Enigma Machine](#)
★ [★ All Tools ★](#)

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```
key: iQZijGdoX0hepv2wnFZ0UsTWU-v6xyGwyqSan_p75CE=
Here's another key for the encryption.
if you are a good cryptographer you can identify the common symmetric encryption.
```

ROUND 3, FIGHT

```
cat key_3.txt
```

```
jlc: 0sPCTS98D3eci4cXumo345l0QiRUS4E17UJQXA0G1uM=
Soi cora ovm zuh pct jee un kitfxwxv hgl gzdglv,
zt xvy tctshe'h hkiehmc kvdu wvs soi jhzyzbf weih nm xys bptysqaioh,
hm xys clgimoamfbr dmkv soi bsxz hfsru'x ncqr xysm avp fdhviomnmeu soi bsxz
```

First guess, Vignere cipher, lets head to dcode.fr again,

The screenshot shows the Dcode.fr website's Vigenere Cipher Decoder. On the left, a search bar contains 'e.g. type scrabble' and a 'GO' button. Below it, the 'Results' section shows 'Vigenere' as a result. The main content area is titled 'VIGENERE CIPHER' and 'VIGENERE DECODER'. It features a 'VIGENERE CIPHERTEXT' input field containing the ciphertext from the previous block. Below this is a 'PARAMETERS' section with dropdowns for 'PLAINTEXT LANGUAGE' (English) and 'ALPHABET' (ABCDEFGHIJKLMNOPQRSTUVWXYZ). A 'DECIPHER METHOD' section has radio buttons for different decryption techniques, with 'KNOWING A PLAINTEXT WORD: KEY' selected. A 'DECIPHER' button is present. To the right, a 'Summary' sidebar lists various links related to Vigenere cipher. At the bottom, there's a 'VIGENERE ENCODER' section with a 'VIGENERE PLAIN TEXT' input field containing 'letme go' and fields for 'CIPHER KEY' (ARTZUMD) and 'ALPHABET' (ABCDEFGHIJKLMNOPQRSTUVWXYZ).

```
key: XeQVPB98P3fve4lJvfk345uARbNDE4F17NFZJBHC1dY=
The last key and you can go decrypte the cipher,
if you couldn't identify then see the starting part of the ciphertext,
if the decryptions with the keys doesn't work then try rearranging the keys.
```

Now this is the part, where things got ugly, apparently the common symmetric encryption used is not AES. Its rather fernet cipher, which could be identified from the head of ciphertext gAAAAA. Moreover, Only the KEY2 was required, rest is just to make you do work 😞

Anyways, we got our flag after so much churning

```
from cryptography.fernet import Fernet
KEY2 = "iQZijGdoX0hepv2wnFZ0UsTWU-v6xyGwyqSan_p75CE="
```

```
f = Fernet(KEY2)
print(f.decrypt(ct))
```

zh3r0-Symm3tric'3ncrypti0n'i5'5tr0ng"

Interactive controls are omitted from this PDF. See the canonical HTML version.